Section A (100 marks)

Question 1 (40 marks)

A silicon-controlled rectifier (SCR), also known as a thyristor, is a semiconductor switching device widely used in medium- and high-power applications. Its ability to control electrical power makes it particularly suitable for AC power control systems.

(a) With aid of waveform, explain the thyristor conduction.

[20 marks, CLO1-PLO1:C2]

(b) Draw the characteristics of the silicon controlled rectifier (thyristor) indicating on your graph the latching current, holding current and peak reverse voltage and the conditions that must be fulfilled to turn on the thyristor.

[10 marks, CLO1-PLO1:C2]

(c) With aid of circuit, explain the apply biasing of thyristor.

[10 marks, CLO1-PLO1:C2]

Question 2 (40 marks)

Power semiconductor devices such as MOSFETs and IGBTs are widely used in power electronics applications due to their efficient switching capabilities. Understanding the differences between these two devices is important for selecting the most suitable component for specific requirement.

(a) Differentiate **Five (5)** characteristic each between MOSFET and IGBT.

[20 marks, CLO2-PLO2:C4]

(b) If a number of p-n junction diodes with identical i-v characteristics are connected in parallel will they share current equally? Justify your answer.

[20 marks, CLO2-PLO2:C4]

Question 3 (20 marks)

Pulse width modulation (PWM) is a modulation technique that generates variable-width pulses to represent the amplitude of an analog input signal. Construct the output of waveform of PWM when the input is V_{tri} (Carrier) and V_{sine} (Reference) and drawing the relevant circuit of PWM.

[20 marks, CLO3-PLO3:C6]

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